MIL-A-17051A(OS)
4 March 1976
SUPERSEDING
MIL-A-17051(Weps)
31 March 1952
(See section 6)

MILITARY SPECIFICATION

AFTERBODY SHELLS, STEEL, WELDED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers welded steel afterbody shells for torpedoes.

*2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATIONS

Military

*MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application

STANDARDS

Military

*MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

*American Society for Testing and Materials (ASTM)

E 8	Tension, Testing of Metallic Materials, Testing of

(Copies are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

*3.1 Preproduction. The afterbody shells furnished under this specification shall be a product which has been inspected and passed the preproduction inspection specified herein.

3.2 Drawing requirements. Afterbody shells shall comply with requirements of the applicable drawings specified in the contract or order (see 6.1).

3.3 Material. Afterbody shells shall be manufactured from extra deep drawing quality sheet steel, full cold-rolled, full pickled, scrubbed, dried, and oiled.

3.3.1 Soundness. Afterbody shell material shall be free from seams, laminations, gas holes, undue segregation or other seriously injurious defects. Specimens shall be examined after macroetching as specified in 4.3.2.2.

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3.3.2 Metallographic structure. The metallographic structure shall be reasonably normal. Excessive abnormalities in grain shape, grain size, carbide distribution, and nonmetallic inclusions shall be cause for rejection.

3.3.3 Chemical composition. Afterbody shells shall comply with the chemical requirements given in table I.

TABLE I. Chemical composition.

Element	Composition (%)	
	Minimum	Maximum
Carbon	0.04	0.16
Manganese	0.20	0.45
Silicon	-	0.30
Phosphorus	-	0.03
Sulphur	-	0.05

3.3.4 Mechanical properties. Mechanical properties of steel in the finished afterbody shell shall comply with the requirements given in table II.

TABLE II. Mechanical properties.

Property	Required value (min.)
Tensile strength, psi	48,000
Elongation in 2 inches, %	18
Hardness, Rockwell B	50

3.3.5 Welded joint.

3.3.5.1 Tensile strength. The tensile strength of material in the welded joint shall be 48,000 pounds per square inch (psi), minimum.

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3.3.5.2 Bend test. Material in the welded joint shall be free from cracks or voids after bending 180° forward or reverse over a 1/2-inch-diameter pin.

3.3.5.3 Soundness of weld. The weld shall completely fill the space between the base metal edges. Fusion of base and deposited metal shall be complete over the entire thickness of the base metal. The weld metal shall be clean, sound, and free from gas holes, unfused edges, dross, or other injurious defects. Specimens shall be examined after macroetching as specified in 4.3.3.3.

3.4 Manufacture.

3.4.1 Construction. Afterbody shells shall be of pieced and welded construction. Not more than two welds shall be used, and they shall be longitudinal only, that is, parallel to the long axis of the afterbody.

3.4.2 Welds. Welds shall be of the open square butt type. Welded bead metal shall completely fill the space and shall be removed smooth and flush with adjacent surfaces both inside and outside of the shell. Neither the welding rod nor flux used shall be such as to cause objectionable corrosive or electrolytic action with the shell material.

3.4.3 Stress relieving. After final welding, afterbody shells shall receive a stress relieving heat treatment of 600° to 650° Fahrenheit (F). for 8 hours.

3.5 Workmanship.

3.5.1 Repair. Afterbody shells shall not be repaired by welding, soldering, or similar methods.

3.5.2 Defects. Afterbody shells shall be clean, smooth, and free from dents, bulges, pits, cracks, rust, scale, and other injurious defects.

3.6 Marking. Afterbody shells shall be permanently identified with the trade mark of the manufacturer. This symbol shall be stamped on the inside of the shell 4-1/2 inches ± 1/2 inch from the large end, and 2 inches ± 1/2 inch from either weld.

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4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

*4.3 Preproduction inspection. Preproduction inspection shall consist of all the inspections of this specification.

*4.3.1 Sampling for preproduction inspection.

*4.3.2 Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish preproduction samples for inspection to determine conformance to the requirements of this specification. Quantity production shall be withheld until the preproduction sample has been pronounced satisfactory by the Government. When a contractor is in continuous production of these items from contract to contract, submission of further preproduction samples may be waived at the discretion of the Government. The approval of preproduction samples or the waiving of the preproduction inspection shall not relieve the contractor of his obligation to submit samples for quality conformance inspection.

4.3.3 Preproduction samples shall consist of two afterbody shells. Samples shall be inspected as specified in the contract or order (see 6.1).

*4.4 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and test specified herein.

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*4.4.1 Sampling for quality conformance inspection. Unless otherwise specified herein and when applicable, sampling plans and procedures used in the determination of the acceptability of products shall be in accordance with the provisions of MIL-STD-105.

4.4.1.1 Lot. A lot shall consist of 50 afterbody shells or a fraction thereof manufactured by the same process and submitted for inspection at the same time.

4.4.1.2 Samples.

4.4.1.2.1 For inspection. Each afterbody shell shall be inspected for requirements of the drawing and this specification.

4.4.1.2.2 For tests. One afterbody shell shall be selected from each lot, and sectioned for test purposes. This sample shell shall be the one with the least perfect weld, as shown by visual examination. At the discretion of the Government, shell blanks may be tested without scrapping the shell by using sections which are normally removed from the blank in subsequent manufacturing operations.

4.5 Test methods.

4.5.1 Afterbody shell material.

*4.5.1.1 Tension test. A section taken from the sample shells shall be rolled flat. A test specimen shall be prepared and tested in accordance with ASTM E 8. No reduction in width near the center shall be made. Test results shall be compared with the requirements shown in table II.

*4.5.1.2 Macroetch. Test pieces, free from all effects of welding shall be taken randomly from the sample shells. The test pieces shall be etched for 10 minutes in a 1 to 1 solution of hydrochloric acid in water at 180° F. The specimens shall be visually examined for compliance with the requirements of 3.3.1.

*4.5.1.3 Metallographic structure. Test pieces, free from all effects of welding shall be taken randomly from the sample shells. The pieces shall be examined at a magnification of X100 for compliance with the requirements of 3.3.2.

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4.5.1.4 Hardness. The specimens shall be tested for Rockwell B hardness and the results compared with table II.

4.5.2 Welded joint. In the following tests of the welded joint, the test specimens shall have the weld centrally located and at right angles to the long axis of the specimen.

4.5.2.1 Tension test. A tension test specimen with the weld in the center of the gage length shall be prepared in accordance with 4.5.1.1. Test results shall be compared with the requirements of 3.3.5.1.

4.5.2.2 Bend test. Two test specimens, 1/2 inch wide and 4 inches long whose cut edges have been ground and corners broken, shall be bent 180° around a 1/2-inch-diameter pin, the bend being centered in the weld. One piece shall be bent forward to place the original outside surface in tension, the other piece shall be bent in reverse to place the original inside surface in tension. The bends shall be examined for compliance with 3.3.5.2.

4.5.2.3 Macroetch. A test piece 1/2 inch wide and 4 inches long whose cut edges have been ground and polished shall be etched for 10 minutes in a 1 to 1 solution of hydrochloric acid in water at 180° F. Should visual examination reveal questionable areas, then a microscopic examination shall be made to determine their extent and nature. The welds shall be examined for compliance with the requirements of 3.3.5.3.

4.6 Retests. If the sample shell selected for tests fails to comply with requirements, an additional sample shell shall be selected from the same lot. Failure of the second shell shall be cause for rejection of the entire lot of 50. Should the second shell be satisfactory, the lot represented shall be provisionally accepted, subject to reimbursement for cost of work performed on shells which later show defects, and such defective shells shall be replaced, or at the discretion of the Government, reimbursement of their cost shall be made.

5. PACKAGING

*5.1 Preservation. Surfaces of afterbody shells shall be coated with corrosion preventive compound conforming to MIL-C-16173, grade I.

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5.2 Packing. Afterbody shells shall be crated or otherwise prepared so as to afford protection from dents and scratches and to insure acceptance by common or other carriers for safe transportation, at the lowest rate, to the point of delivery.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

*6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Applicable drawing numbers

c. Whether preproduction inspection is required (see 4.1.1.3)

d. Where the preproduction inspection is to be conducted (by contractor or by the Government) (see 4.1.1.4).

*6.2 Supersession data. This specification includes the requirements of NAVORD OS No. 886 dated 24 April 1944.

*6.3 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-A-17051A(OS) Afterbody Shells, Steel, Welded

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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